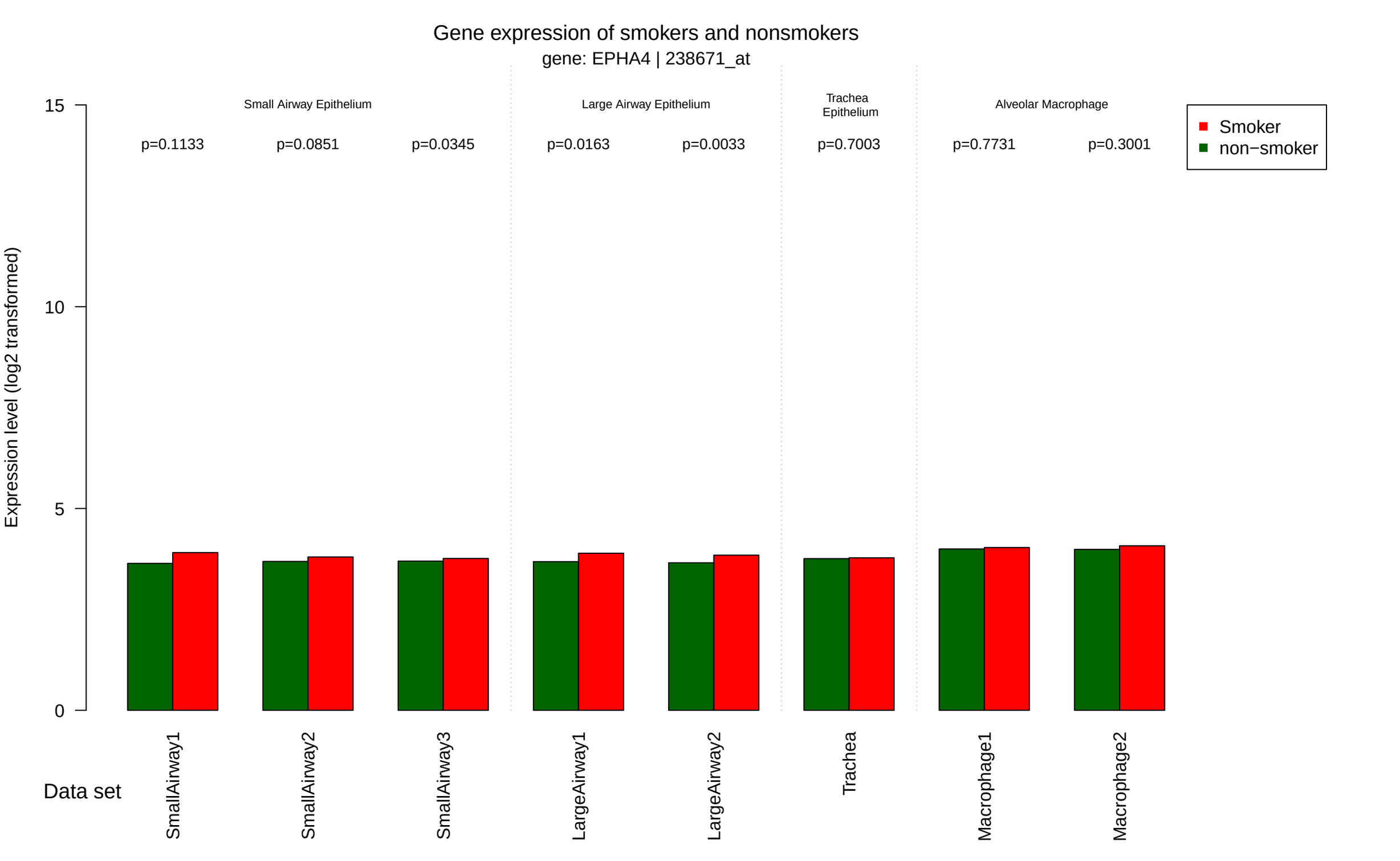
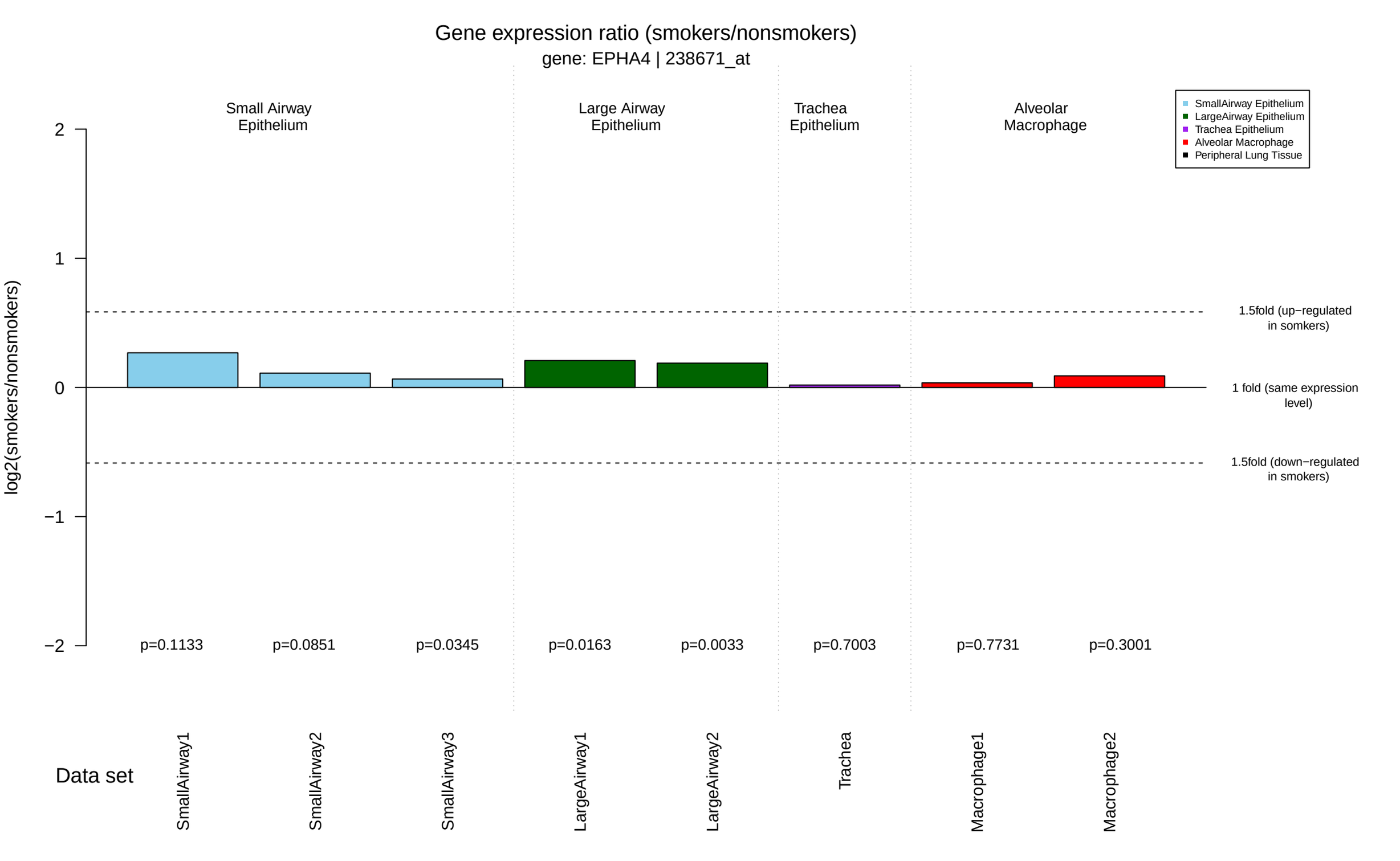
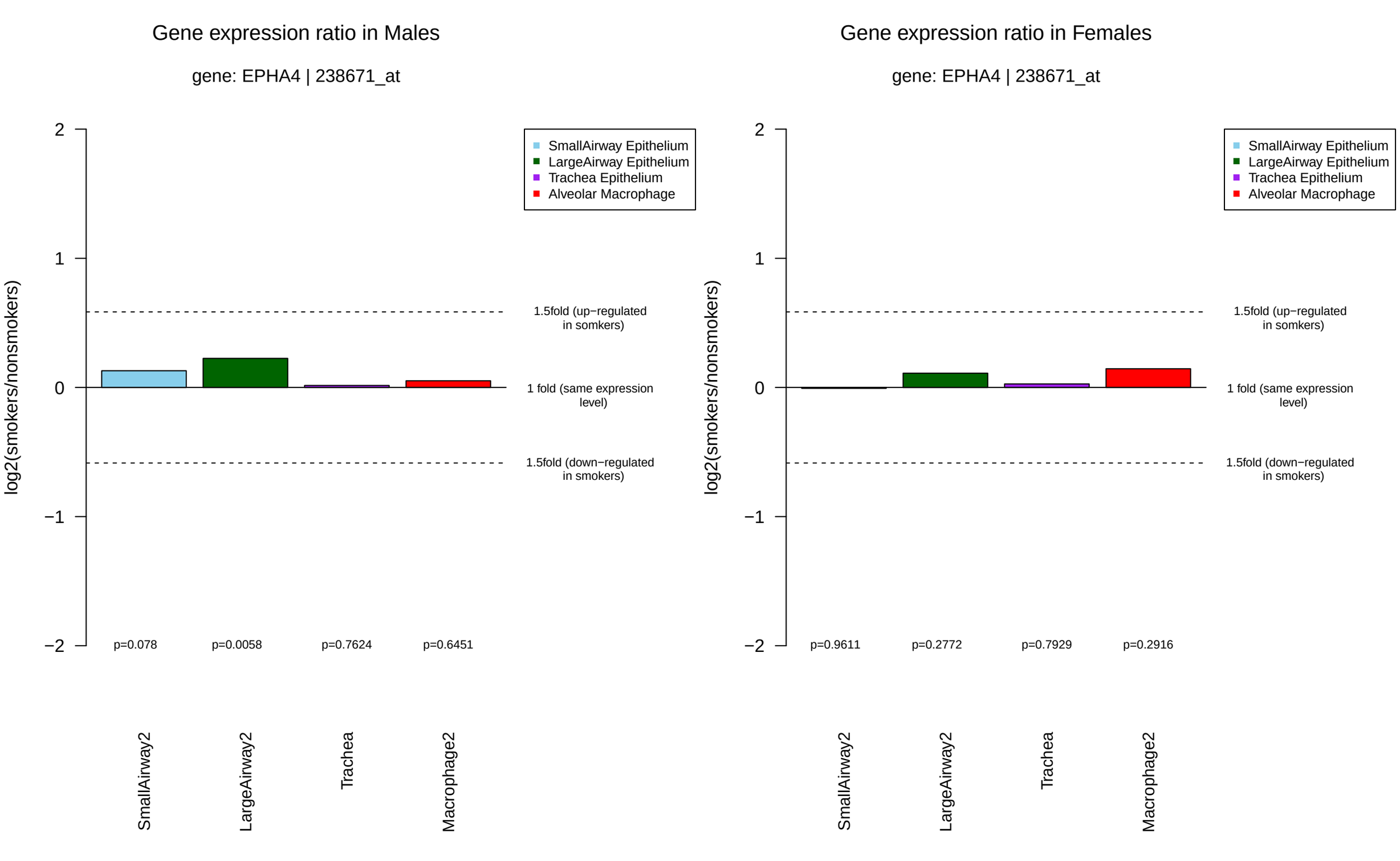
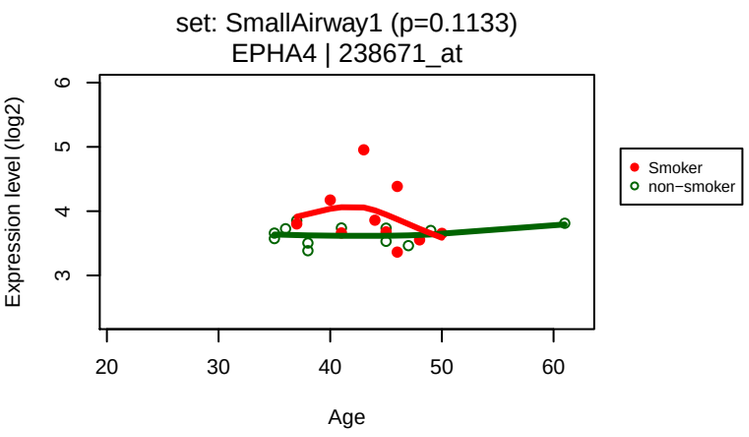
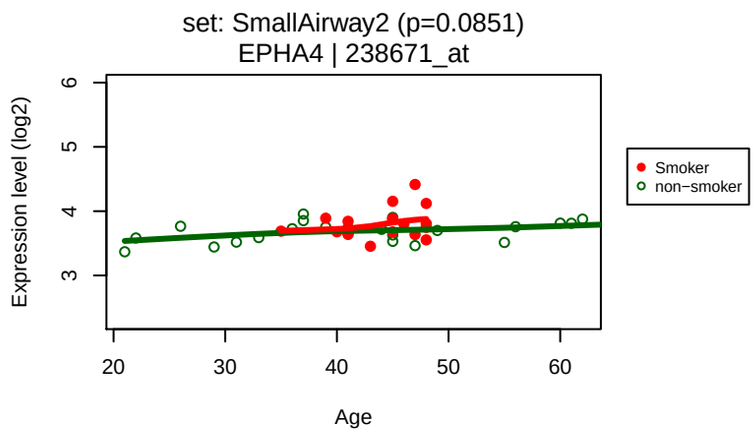
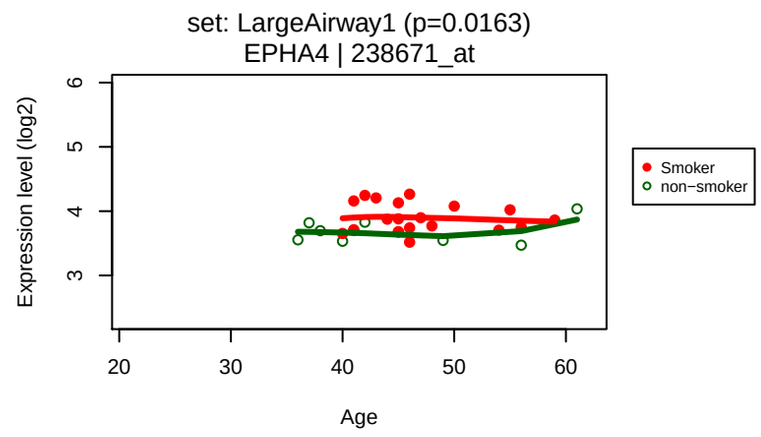

Gene expression of smokers and nonsmokers
gene: EPHA4 | 238671_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1133
p=0.0851
p=0.0345
p=0.0163
p=0.0033
p=0.7003
p=0.7731
p=0.3001
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

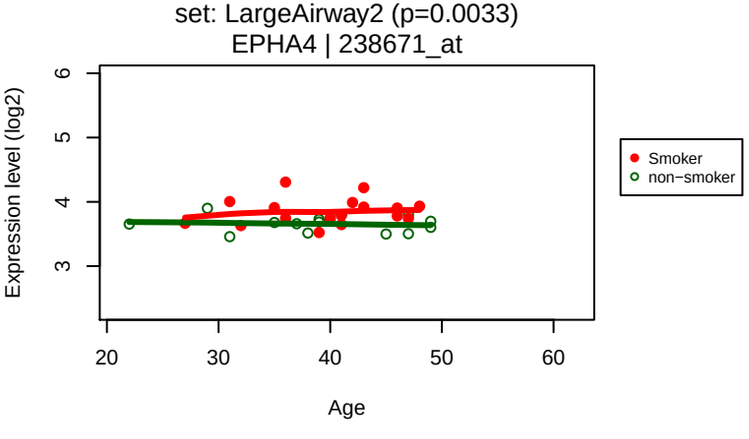

Gene expression ratio (smokers/nonsmokers)
gene: EPHA4 | 238671_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1133
p=0.0851
p=0.0345
p=0.0163
p=0.0033
p=0.7003
p=0.7731
p=0.3001
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

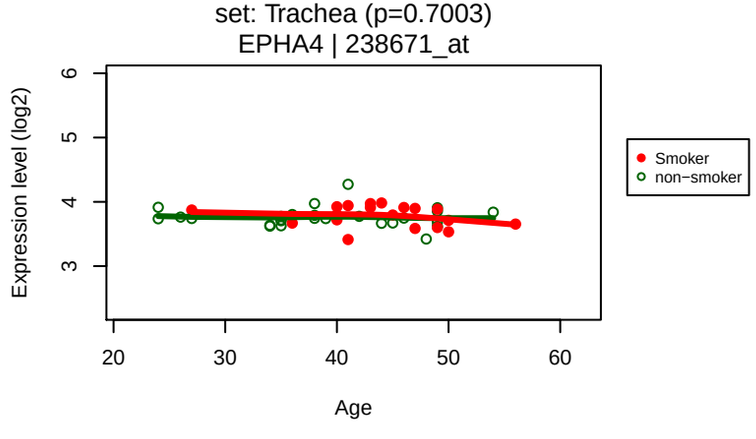

Gene expression ratio in Males
gene: EPHA4 | 238671_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.078
p=0.0058
p=0.7624
p=0.6451
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: EPHA4 | 238671_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9611
p=0.2772
p=0.7929
p=0.2916
SmallAirway2
LargeAirway2
Trachea
Macrophage2

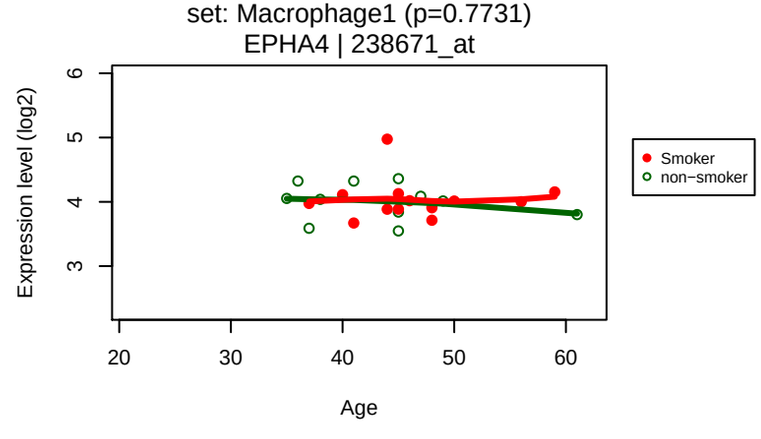

set: SmallAirway1 (p=0.1133)
EPHA4 | 238671_at
Expression level (log2)
Age
Smoker
non−smoker

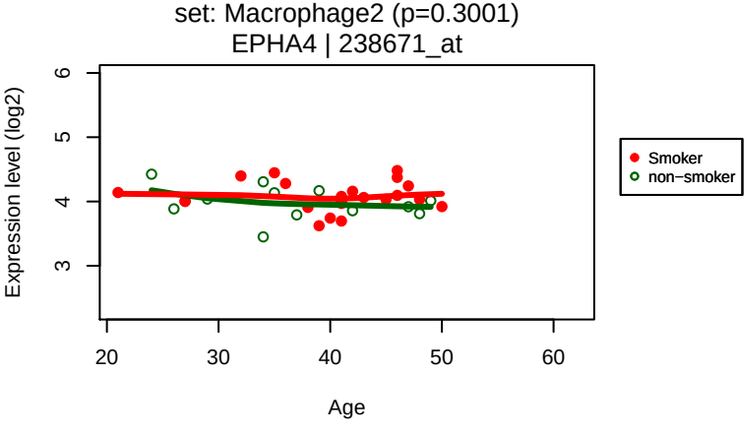

set: SmallAirway2 (p=0.0851)
EPHA4 | 238671_at
Expression level (log2)
Age
Smoker
non−smoker

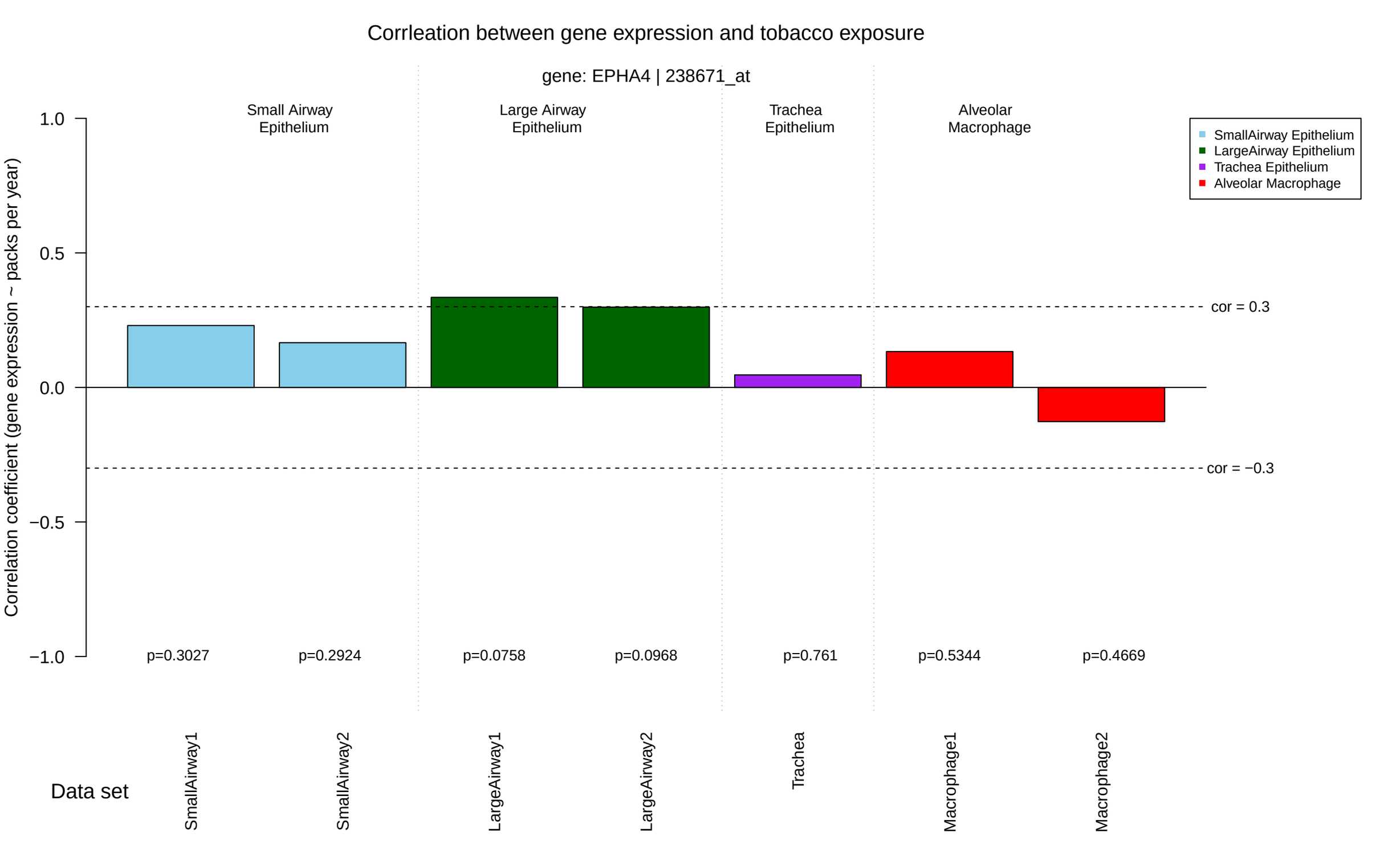

set: LargeAirway1 (p=0.0163)
EPHA4 | 238671_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0033)
EPHA4 | 238671_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7003)
EPHA4 | 238671_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7731)
EPHA4 | 238671_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3001)
EPHA4 | 238671_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: EPHA4 | 238671_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3027
p=0.2924
p=0.0758
p=0.0968
p=0.761
p=0.5344
p=0.4669
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2